



**NANYANG
TECHNOLOGICAL
UNIVERSITY**
SINGAPORE

SC3021 – Data Science Fundamentals

Lab 02: Justification of Source Selection and Data Preparation

Group 3, SDAC

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Lab 2: Data Science Process



PREPARE Stage:

1. Source Identification
2. Source Exploration
3. Source Justification

PROCESS Stage:

1. Source Preparation

Prepare (B): Source Identification, Exploration

- DS1: Traditional Learning Behaviour (Primary Dataset)
 - Study hours per day and study consistency.
 - Class attendance and class participation.
 - Assignment scores, exam scores, and final academic performance.
- DS2: Student Lifestyles
 - Social media usage and Netflix viewing hours.
 - Part-time job status and extracurricular participation.
- DS3: Usage of AI in Studies of AI in their studies
 - AI tools – AI usage, Name of AI tool, Percentage of AI generated content.
 - Class Attendance and Participation.
 - Study hours, exam scores, and other learning behavior.

When we try to move to the “process” stage, we realize that these three datasets couldn’t enrich each other.

Each datasets is taken from different subset of student, and hence difficult to enrich one another. Hence, we decided to change our datasets.

Source Identification

- ~~DS1: Traditional Learning Behaviour (Primary Dataset)~~
 - ~~Study hours per day and study consistency.~~
 - ~~Class attendance and class participation.~~
 - ~~Assignment scores, exam scores, and final academic performance.~~
- DS2: Student Lifestyles
 - Social media usage and Netflix viewing hours.
 - Part-time job status and extracurricular participation.
- **DS3 (NEW DS1): Usage of AI in Studies of AI in their studies**
Study Behavior and Academic Performance
 - AI tools – AI usage, Name of AI tool, Percentage of AI generated content.
 - Class Attendance and Participation.
 - Study hours, exam scores, and other learning behavior.

DS1 and DS3 **shares a lot of common features**, with DS3 having a more features (which includes AI Usage), thus we chose to drop DS1 and replace with DS3 as the **main datasets**.

We keep DS2, as they have some “Lifestyle” features that we hope we can “transfer” to DS3.

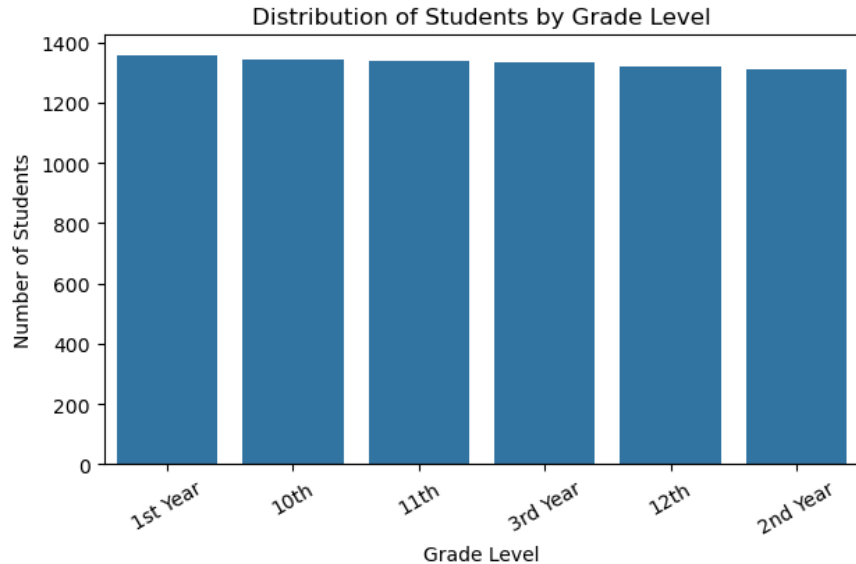
DS1: Learning Behaviour (Main Dataset)

Features present in DS1:

- **Student Demographics**
 - student_id
 - age
 - gender
 - grade_level
- **Study Behaviour and Learning Habits**
 - study_hours_per_day
 - study_consistency_index
 - tutoring_hours
 - class_participation_score
- **Lifestyle and Daily Behaviour**
 - sleep_hours
 - social_media_hours
- **AI Usage in Learning**
 - uses_ai
 - ai_usage_time_minutes
 - ai_tools_used
 - ai_usage_purpose
 - ai_dependency_score
 - ai_generated_content_percentage
 - ai_prompts_per_week
 - ai_ethics_score
- **Academic Performance Indicators**
 - last_exam_score
 - assignment_scores_avg
 - attendance_percentage
 - concept_understanding_score
 - improvement_rate
 - final_score

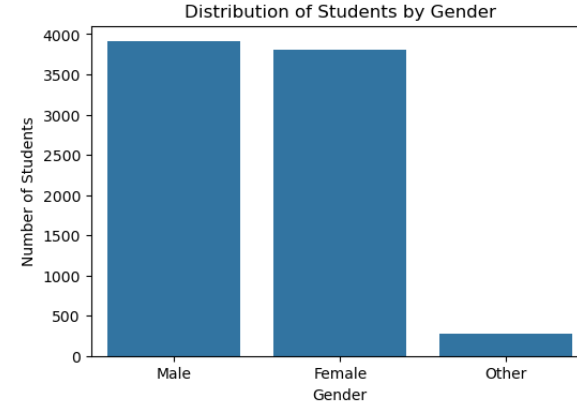
DS1: Profiling – Exploring Distributions

Distribution of Students Across Grade Levels

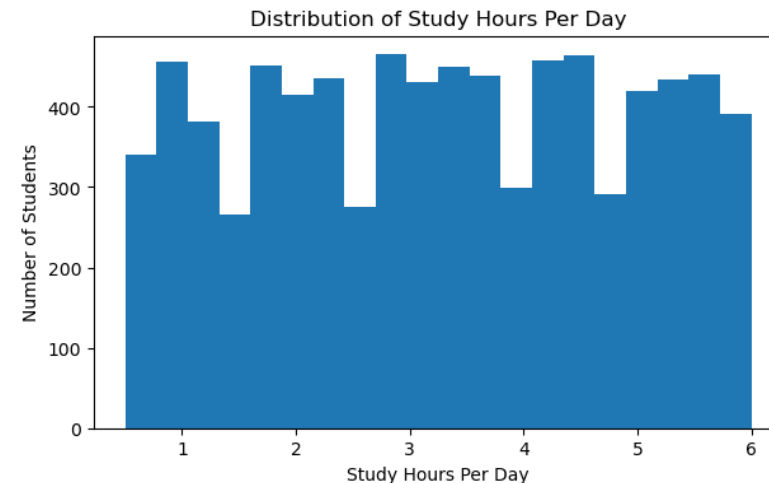


- Dataset spans multiple education levels → broader view of study habits, AI usage, and performance
- Learning behaviors differ across levels (e.g., school vs university)
 - University students have more autonomy in studying
- Grade level is important in explaining differences in behaviour and outcomes

Distribution of Students By Gender



Distribution of Students By Study Hours



DS1: Data Quality and Structural Profiling

We conduct **data quality** and **structural profiling** on DS1. This is to assess whether the dataset is suitable for modelling and to identify preprocessing issues.

- Checked **missing values** → identify need for imputation/removal
- Removed **duplicate records**
- Profiled **categorical variables** → detect inconsistencies, imbalance, encoding needs
- Analyzed **numeric correlations (heatmap)** → identify relationships, multicollinearity, key predictors

DS1: Data Quality and Structural Profiling

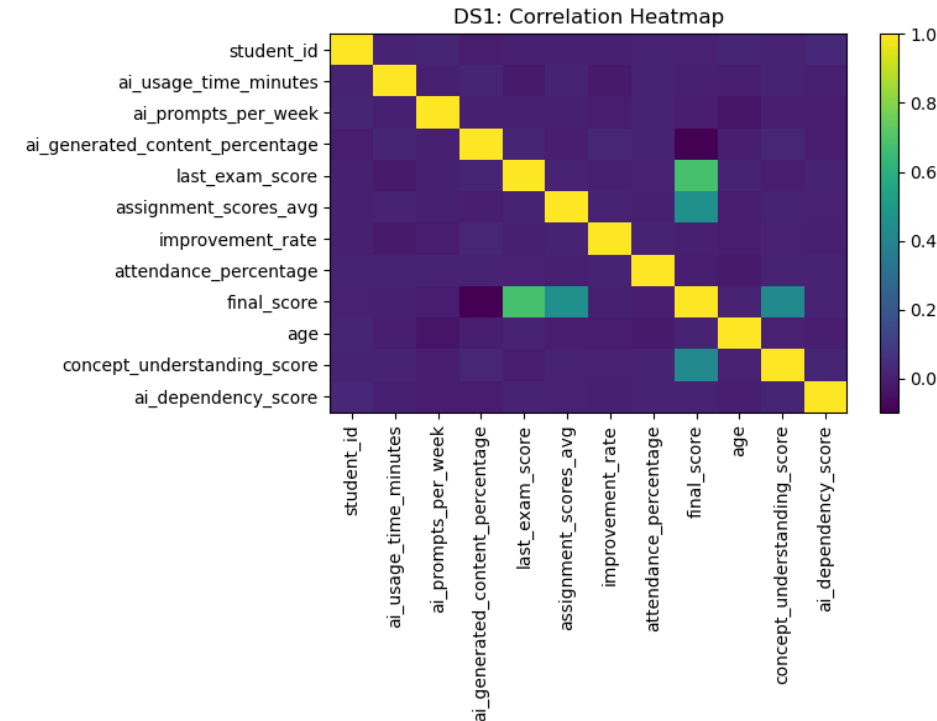
Missing Values

	missing_count	missing_pct
ai_tools_used	1362	17.025
ai_usage_purpose	1346	16.825
student_id	0	0.000
assignment_scores_avg	0	0.000
passed	0	0.000
final_score	0	0.000
class_participation_score	0	0.000
tutoring_hours	0	0.000
social_media_hours	0	0.000
sleep_hours	0	0.000

Categorical Profiling

Column: gender unique_count = 3			Column: grade_level unique_count = 6		
percentage			percentage		
gender			grade_level		
Male	48.8500		1st Year	16.9500	
Female	47.6375		10th	16.8000	
Other	3.5125		11th	16.7000	
			3rd Year	16.6750	
Column: ai_tools_used unique_count = 5			12th	16.5125	
percentage			Column: ai_usage_purpose unique_count = 5		
ai_tools_used			percentage		
ChatGPT+Gemini	21.075625		ai_usage_purpose		
ChatGPT	20.066285		Exam Prep	20.559062	
Gemini	19.749925		Doubt Solving	20.168320	
Claude	19.734860		Homework	20.018034	
Copilot	19.373305		Coding	19.762549	
			Notes	19.492035	

Correlation Heatmap



Duplicate Values

DS1 – Duplicates

```
=====
{'duplicate_rows': np.int64(0),
 'duplicate_percentage': np.float64(0.0)}
```


DS1: Quality Evaluation

- Completeness
 - Rich set of variables → suitable for behavioral analysis.
- Interpretability
 - Clear, understandable features (e.g., study hours, attendance, AI usage).
- Dataset Reputation
 - From Kaggle → widely used but unclear data provenance.
- Overall: **Suitable as core dataset despite limitations.**

Recap: Source Identification

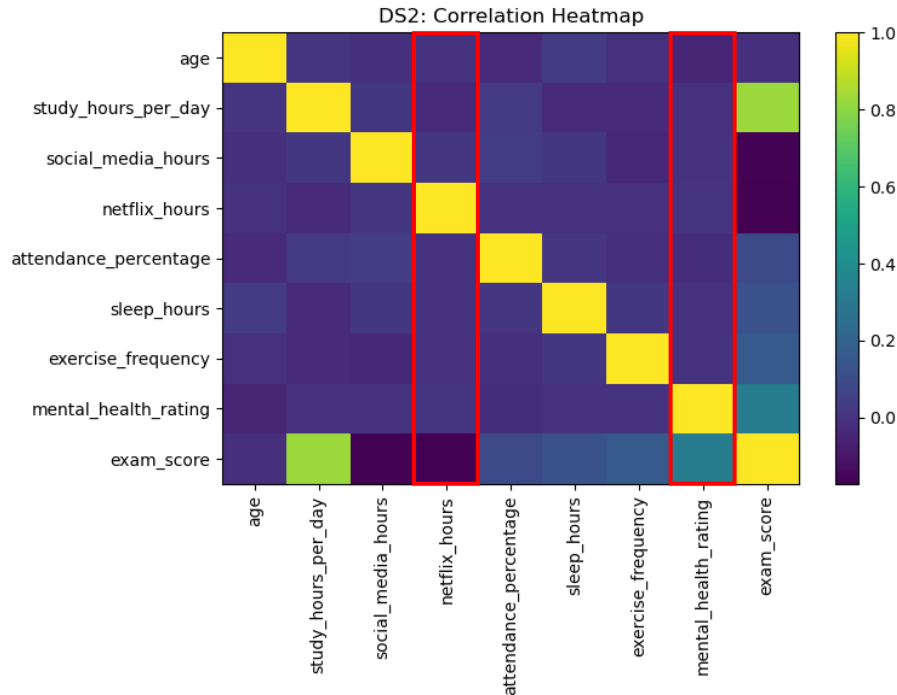
- DS1: Study Behavior and Academic Performance
 - AI tools – AI usage, Name of AI tool, Percentage of AI generated content.
 - Class Attendance and Participation.
 - Study hours, exam scores, and other learning behavior.
- DS2: Student Lifestyles
 - Social media usage and Netflix viewing hours.
 - Part-time job status and extracurricular participation.

Observations and Rationale:

- Some lifestyle factors (e.g., sleep, social media) already appear and overlaps in DS1
 - However, DS1 lacks deeper psychological and wellbeing variables (e.g., mental health, lifestyle quality)
 - Rationale: Use DS2 to enrich DS1, leveraging overlapping features as linkage points for integration
- This allows us to incorporate additional dimensions: social media, Netflix, sleep, exercise, diet, internet quality, part-time job, extracurriculars, mental health

DS2: Correlation Analysis

To be able to do **transfer learning**, there should be a **correlation** between the **overlapping factors** (sleep hours, study hours per day, social media hours, age, gender, attendance) with the **features we want to transfer** (mental health rating, Netflix hours, etc).



Linear Regression Performance

R2 : -0.0034
RMSE : 2.6555
MAE : 2.2570

Feature Coefficients

	feature	coefficient
0	age	-0.046073
3	sleep_hours	-0.043021
1	study_hours_per_day	-0.028295
2	social_media_hours	-0.024889

We also validated the relationship using a **linear regression** model. The results show a **very low R^2** , indicating that the **model explains only a small portion of the variance in the data**. This means the **prediction error is high**, and there is little to **no meaningful linear relationship** between the variables.

Small/NO correlation between mental health score and other behavioral variables.

Recap: Source Identification

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- DS2: Student Lifestyles
 - Social media usage and Netflix viewing hours.
 - Part-time job status and extracurricular participation.
- DS4: Student Psychological Wellbeing
 - Lifestyle Factors (Sleep Hours, Study Hours, etc)
 - Wellbeing index, Burnout risk, Anxiety score, overthinking score

Rationale:

- We try to find other datasets **similar to DS2**, which have **similar lifestyle factors** (Sleep hours, study hours, age, gender, screen time hours).
- Also have **Social Factors** (wellbeing index, burnout risk, motivation level, etc.)
- But **can be used for transfer learning** (High correlation between transferred variables)

DS4: Student Psychological Wellbeing

Features present in DS4:

- **Student Demographics**

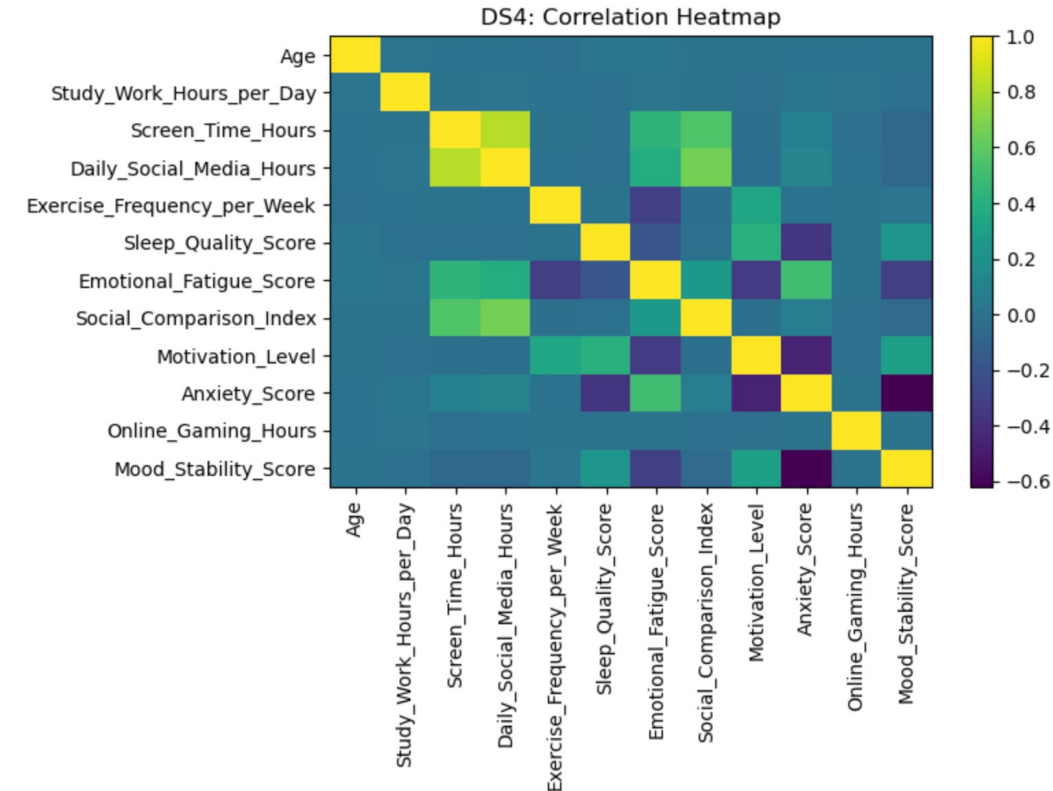
- age
- gender
- country

- **Lifestyle and Daily Behaviour**

- student_working_hours
- daily_social_media_hours
- screen_time_hours
- night_scrolling_frequency
- online_gaming_hours
- content_type_preference
- exercise_frequency_per_week
- study_work_hours_per_day

- **Behavioural Indicators**

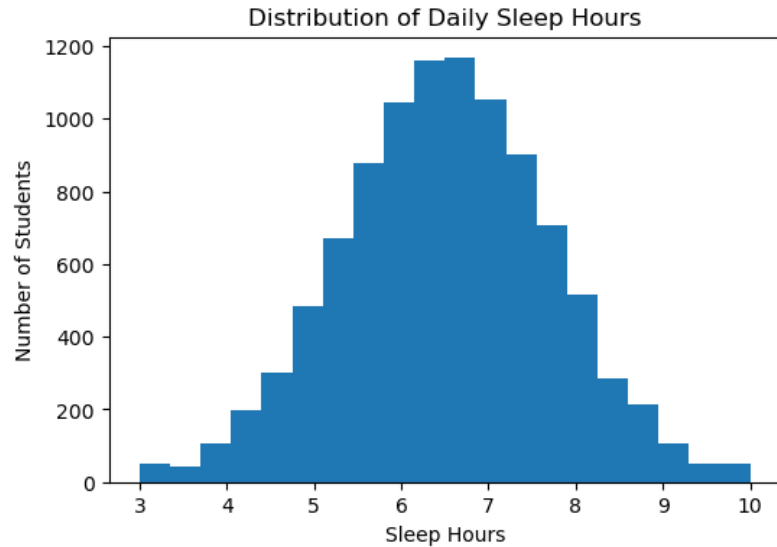
- overthinking_hours
- anxiety_score
- mood_stability_score
- social_comparison_index
- sleep_quality_score
- motivation_level
- emotional_fatigue_score
- wellbeing_index
- burnout_risk



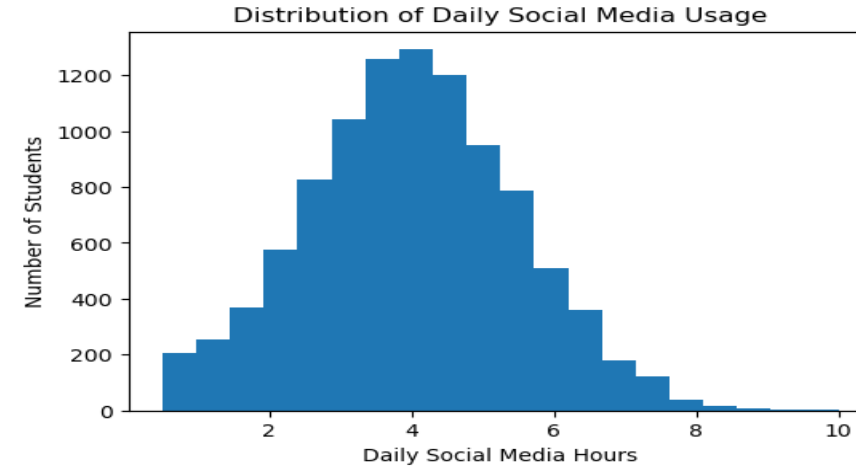
After analyzing the correlation heatmap, turns out there is **relationship** between the **common feature** and **feature we want to transfer**.

DS4: Profiling – Exploring Distributions

Distribution of Sleep Hours Across Students

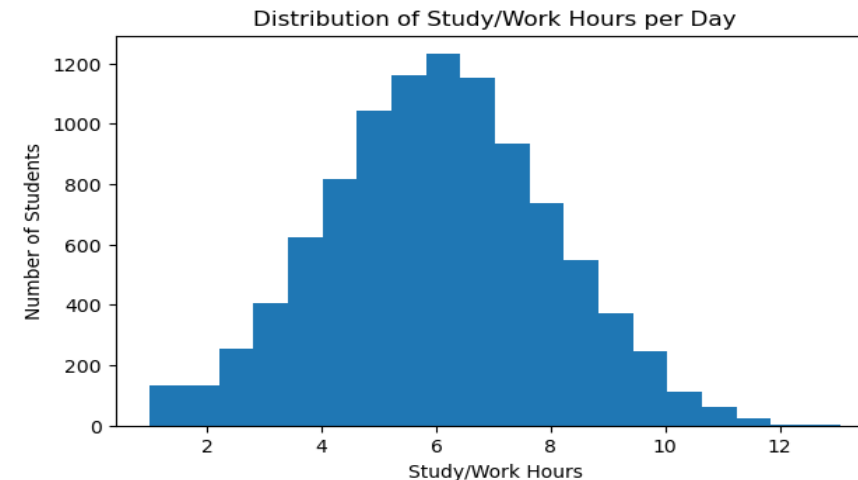


Distribution of Daily Social Media Usage Across Students



- Important factors like sleep hours and social media usage affect psychological wellbeing and burnout risk
- **Variability** in these different factors across the students

Distribution of Study-Work Hours Across Students



DS4: Profiling – Numeric & Categorical

- No missing values
- Reasonable variability across important factors, suitable for predictive modelling
- Potential class imbalance in burnout risk

	mean	std	min	25%	median	75%	max
Age	21.992900	2.589309	18.00	20.0000	22.00	24.0000	26.00
Anxiety_Score	4.747556	1.305593	1.00	3.8400	4.74	5.6300	9.69
Caffeine_Intake_Cups	1.505871	0.934069	0.00	0.8000	1.48	2.1500	6.00
Daily_Sleep_Hours	6.512031	1.194194	3.00	5.7100	6.52	7.3200	10.00
Daily_Social_Media_Hours	4.005883	1.480509	0.50	3.0100	3.99	4.9825	10.00
Emotional_Fatigue_Score	5.582222	1.416558	1.00	4.6200	5.58	6.5400	10.00
Exercise_Frequency_per_Week	2.994855	1.478248	0.00	1.9600	3.01	4.0000	7.00
Mood_Stability_Score	5.161021	1.257055	1.00	4.3100	5.17	6.0000	10.00
Motivation_Level	5.086132	1.308351	1.00	4.2000	5.08	5.9500	10.00
Night_Scrolling_Frequency	2.986009	1.197035	0.00	2.1875	2.98	3.7900	7.00
Online_Gaming_Hours	1.607241	1.300045	0.00	0.4700	1.46	2.5000	6.00
Overthinking_Score	5.001868	1.143179	1.00	4.2400	5.00	5.7600	9.51
Screen_Time_Hours	7.016991	1.792529	2.00	5.8100	7.00	8.2100	14.00
Sleep_Quality_Score	6.514447	1.427806	1.02	5.5500	6.53	7.4900	10.00
Social_Comparison_Index	4.403556	1.336479	1.00	3.4700	4.40	5.3200	9.42
Study_Work_Hours_per_Day	6.030983	1.975337	1.00	4.7000	6.03	7.3400	13.05
Wellbeing_Index	3.870213	1.238940	1.00	3.0200	3.87	4.7000	8.86

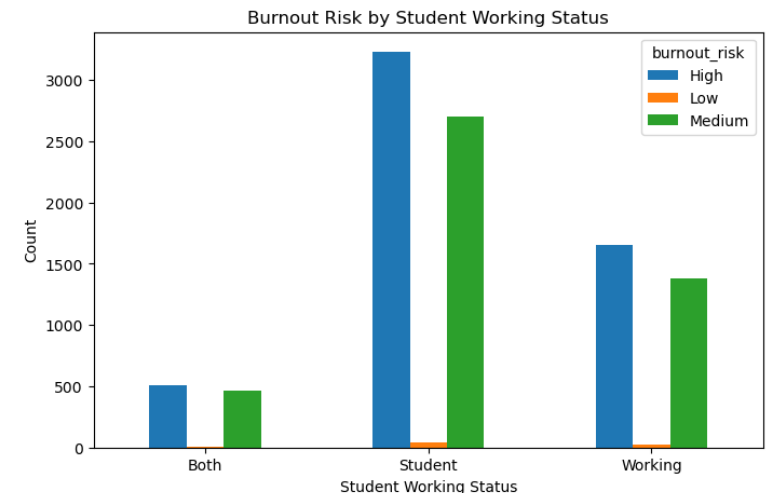
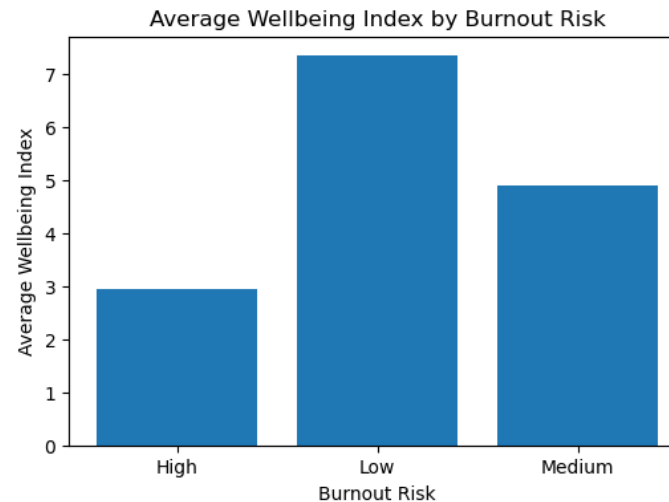
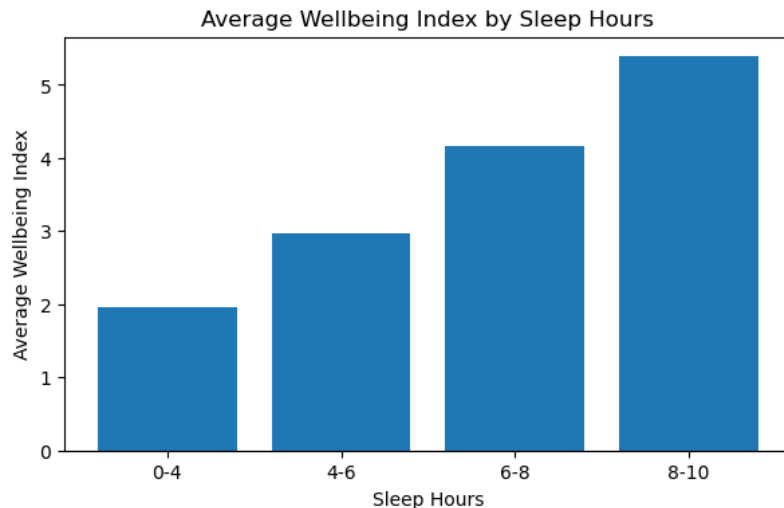
Numeric Profile

percentage		percentage	
Gender		Burnout_Risk	
Male	45.47	High	53.88
Female	44.81	Medium	45.48
Non-binary	9.72	Low	0.64
percentage			
Student_Working_Status			
Student		59.71	
Working		30.56	
Both		9.73	

Categorical Profile

DS4: Considerations

- **No common identifier** in students (similar to DS2, cannot merge with DS1)
- Source for **enriching DS1** with **psychological wellbeing and burnout risk indicators**.
- Standardised some column names with that of DS1 to facilitate transfer learning
- Generated some additional plots which indicated **positive correlation between sleep hours and wellbeing & negative correlation between burnout risk and wellbeing**. Students that were **both working and studying** also had **high burnout risk** compared to doing only either



Recap: Source Identification

- DS1: Study Behavior and Academic Performance
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- DS4: Student Psychological Wellbeing
 - Lifestyle Factors (Sleep Hours, Study Hours, etc)
 - Wellbeing index, Burnout risk, Anxiety score, overthinking score
- DS3: Social Media Usage and Addiction Behavior
 - Daily social media usage, Most used platform
 - Sleep hours per night, Mental health score
 - Social media addiction score

Rationale:

- We try to find other datasets that **provides more specific information on social media addiction.**
- This may directly influence students' concentration, productivity, and study outcomes.
- Therefore, **DS3 is a suitable candidate** dataset for **transfer learning**, allowing us to estimate students' social media addiction levels and enrich the DS1 dataset.

DS3: Data Quality and Structural Profiling

Similar to DS1, we perform **data quality** and **structural profiling**.

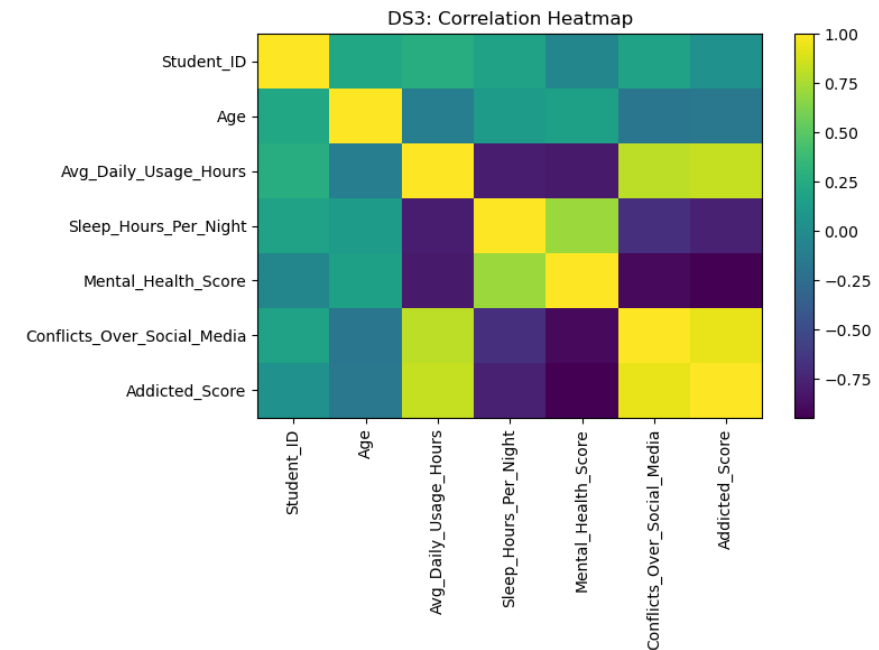
Missing Values

	missing_count	missing_pct
Student_ID	0	0.0
Age	0	0.0
Gender	0	0.0
Academic_Level	0	0.0
Country	0	0.0
Avg_Daily_Usage_Hours	0	0.0
Most_Used_Platform	0	0.0
Affects_Academic_Performance	0	0.0
Sleep_Hours_Per_Night	0	0.0
Mental_Health_Score	0	0.0
Relationship_Status	0	0.0
Conflicts_Over_Social_Media	0	0.0
Addicted_Score	0	0.0

Categorical Profiling

Column: Gender unique_count = 2		Column: Academic_Level unique_count = 3	
percentage		percentage	
Gender		Academic_Level	
Female	50.070922	Undergraduate	50.070922
Male	49.929078	Graduate	46.099291
		High School	3.829787
Column: Country unique_count = 110		Column: Most_Used_Platform unique_count = 12	
percentage		percentage	
Country		Most_Used_Platform	
India	7.517730	Instagram	35.319149
USA	5.673759	TikTok	21.843972
Canada	4.822695	Facebook	17.446809
Denmark	3.829787	WhatsApp	7.659574
Ireland	3.829787	Twitter	4.255319
Column: Affects_Academic_Performance unique_count = 2			
percentage			
Affects_Academic_Performance			
Yes	64.255319		
No	35.744681		

Correlation Heatmap



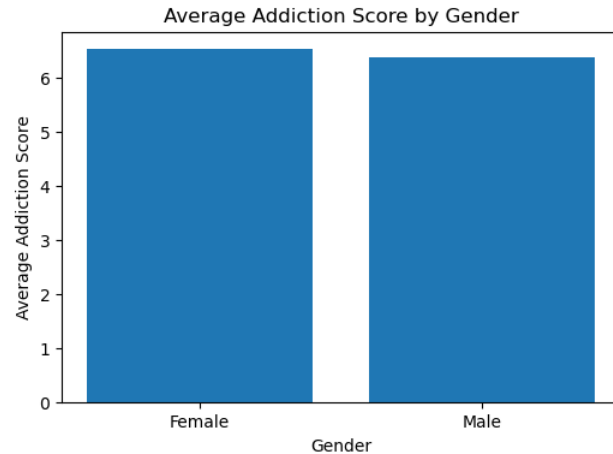
Duplicate Values

DS3 – Duplicates

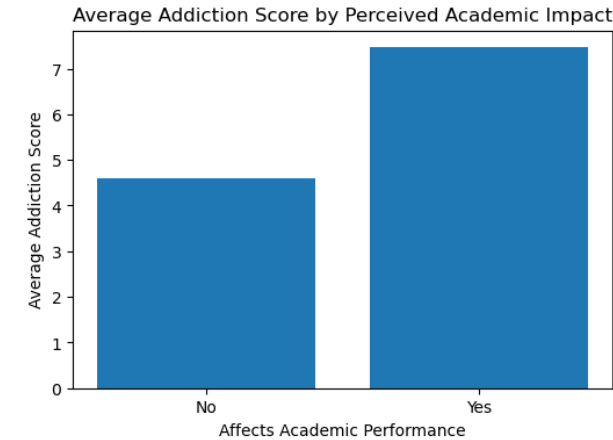
```
=====
{'duplicate_rows': np.int64(0),
 'duplicate_percentage': np.float64(0.0)}
```


DS3: Profiling – Exploring Distributions

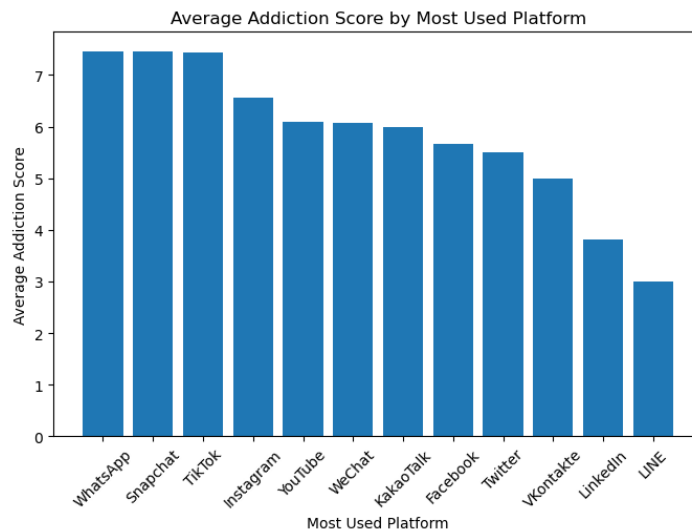
Distribution of Addiction by Gender



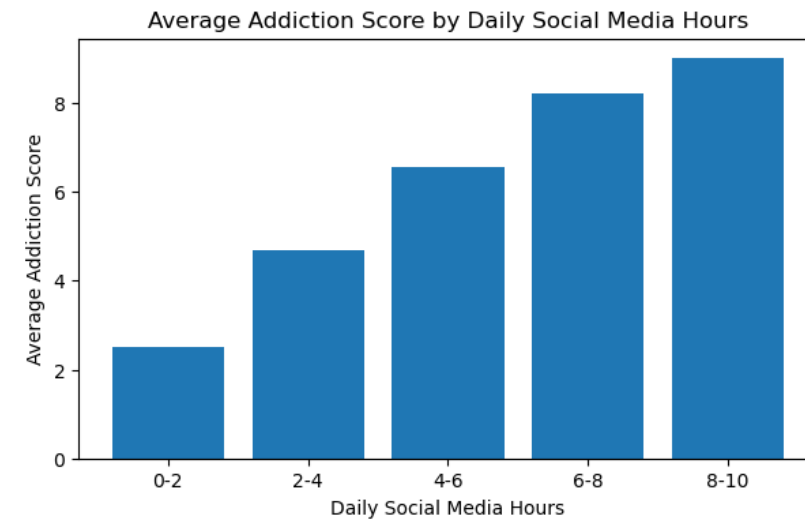
Distribution of Students By Perceived Academic Impact



Distribution of Addiction by Most Used Platform



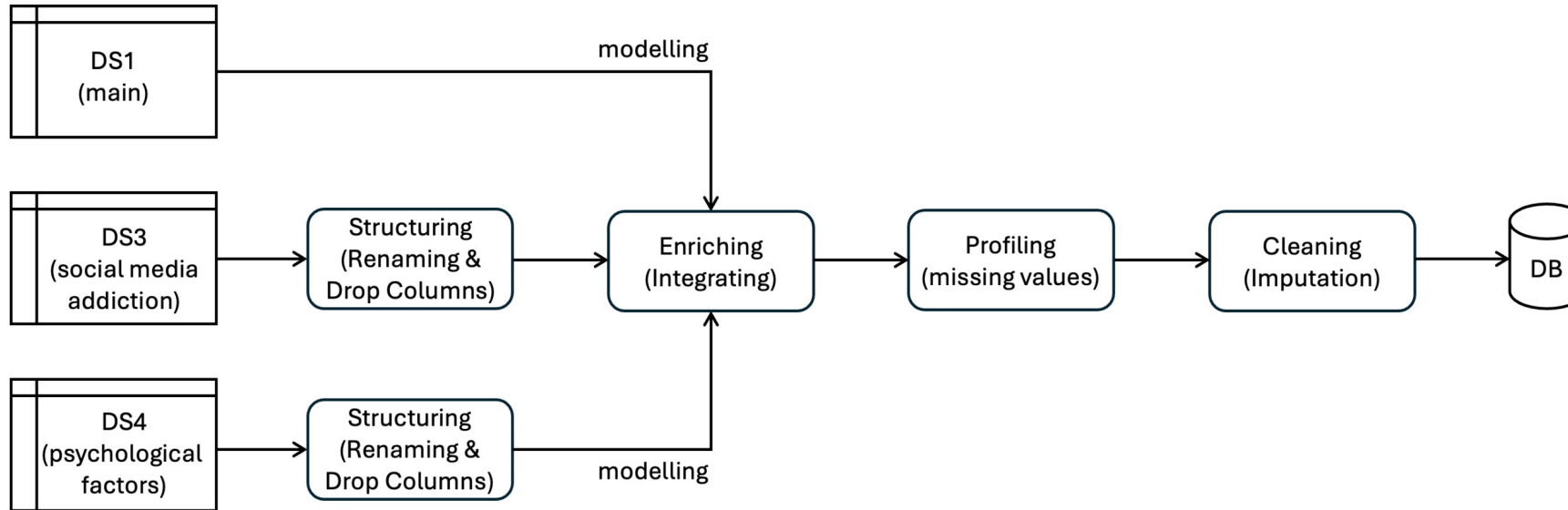
Distribution of Students By Study Hours



DS3: Structuring and Data Quality

- To be able to do **transfer learning**, we need to do structuring as well, which is *renaming column*.
 - Standardized **column names in DS3 to match DS1**
 - Align common variables to enable **easy integration across datasets**
 - Ensures consistent structure for **feature transfer and modelling**.
- Conclusion:
 - DS3 provides meaningful social media behavioural signals
 - Includes a strong **outcome variable (addicted_score)** with clear relationships to predictors
 - Models show addiction can be predicted reliably using features shared with DS1
 - **Used for transfer learning to enrich DS1 with estimated addiction scores**
 - Adds behavioural context to improve academic performance prediction

PROCESS: Data Preparation Overview



- Retained relevant and necessary attributes from DS1, DS3 & DS4 (structuring & profiling)
- Decided on a transfer learning approach to enrich DS1 (main dataset)
 - DS3 to estimate social media addiction levels
 - DS4 to model psychological wellbeing and burnout risk
- Further profiling to check for missing values and quality
- Handling missing values through imputation (data cleaning)
- Resulting dataset used for prediction model for students' academic performance.

DS3 Structuring & Transferring to DS1

- To **enrich DS1** with a **predicted social media addiction** score
- Used *age, gender, social_media_hours, sleep_hours* to output a target variable, *addicted_score*.
- **Modelling Preprocessing**
 - Create pipeline for Initial imputation to ensure data is clean and suitable for machine learning models.
 - Split DS3 into **training sets** to fit the models and **testing sets** to evaluate the models.

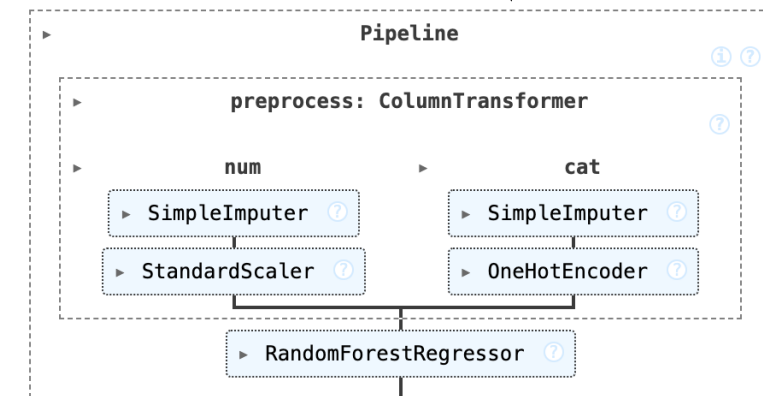
- **Modelling**

- Evaluated several regression models to predict *addicted_score* using **RMSE, MAE, R²**

	model	RMSE	MAE	R2
0	RandomForest(n=400)	0.786347	0.548563	0.783463
1	GradientBoosting	0.838104	0.632636	0.754020
2	Lasso(alpha=0.001)	0.921929	0.742592	0.702355
3	LinearRegression	0.922615	0.743078	0.701912
4	Ridge(alpha=1.0)	0.922647	0.743339	0.701891

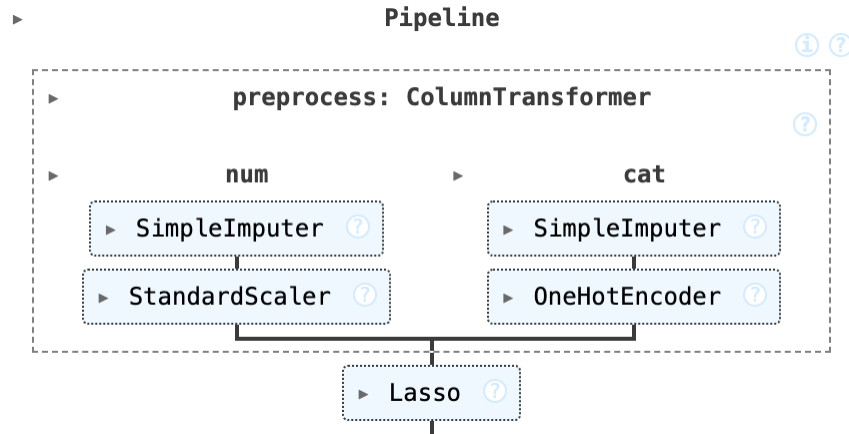
- **Post-Processing and Transfer**

- Retrain model on entire DS3
- Applied on DS1 to generate new variable: *predicted_social_media_addicted_score*



DS4 Structuring & Transferring to DS1

- To **enrich DS1** with **psychological indicators**
- Used *age, gender, social_media_hours, sleep_hours & study_work hours* to output 2 target variables, *predicted_wellbeing, predicted_burnout_risk*.
- **Modelling Preprocessing**
 - Create pipeline for Initial imputation to ensure data is clean and suitable for machine learning models.
 - Numeric variables imputed with median & standardized
 - Categorical variable gender imputed with most frequent value and one-hot encoded



- Split DS4 into **training sets** to fit the models and **testing sets** to evaluate the models.

DS4 Structuring & Transferring to DS1

- **Modelling**

- **Regression task** to derive continuous variable *predicted_wellbeing*.

- Evaluated models similarly using RMSE, MAE, R²

	model	RMSE	MAE	R2
0	Lasso(alpha=0.001)	0.873941	0.700831	0.507676
1	LinearRegression	0.874025	0.700897	0.507582
2	Ridge(alpha=1.0)	0.874026	0.700898	0.507580
3	GradientBoosting	0.875544	0.701363	0.505868
4	RandomForest(n=400)	0.909056	0.729098	0.467318

- **Classification task** to derive categorical variable *predicted_burnout_risk*.

	model	Accuracy
0	LogisticRegression	0.7525
1	GradientBoostingClassifier	0.7525
2	RandomForestClassifier(n=400)	0.7355

- **Post-Processing and Transfer**

- Retrain model on entire DS4
- Applied on DS4 to generate new variable: *predicted_wellbeing* and *predicted_burnout_risk*

Final Dataset Analysis

- Enriched DS1 now contains **additional behavioral features**
- Final **missing value analysis** showed only **ai_tools_used** and **ai_usage purpose** to have missing values
- Removing rows with missing values would result in unnecessary bias and reduction of dataset size.

Engineered variables through other datasets

student_id	age	...	study_work_hours	predicted_wellbeing	predicted_social_media_addicted_score	predicted_burnout_risk
1	20	...	7.1	4.640467	2.905	Medium
2	17	...	6.1	4.545657	5.510833	Medium
3	24	...	4.8	3.676014	5.27	High
4	21	...	7.3	5.078283	5.349583	Medium
5	18	...	6.4	5.875005	2.63	Medium
...	...	...	...	...	...	...
7996	23	...	4	5.390904	2.8625	Medium
7997	21	...	6.2	3.066709	5.3175	High
7998	24	...	1.7	2.835672	5.425	High
7999	22	...	5.9	2.362419	8.046667	High
8000	15	...	6.7	5.795682	4.42	Medium

Combining attributes
(study_hours + tutoring hours)

Final Dataset Analysis: Numerical Profiling

- **Student Demographics:** Aged 16-22 (~ 19 years old), typically upper sec/tertiary.
- **Study Behavior**
 - Study ~ 3.29 hours/day,
 - Combined study-work hours ~ 5.81 hours/day
 - IQR (4.3-7.3hrs) suggest moderate variance in study commitment.
- **AI Usage Behavior**
 - 64% of students use AI Students spend ~ 89mins/day using AI
 - AI dependency score ~ 5.5/10
- **Academic Performance:** Final score ~ 56.8, most between 47.3 - 66.2, pass rate 88.9%
- **Lifestyle Behavior**
 - Students sleep ~ 6.47hrs/night
 - Social media usage ~ 3hrs/day, with some reaching up to 6
- **Psychological Indicators:**
 - Predicted wellbeing score ~ 3.95 (moderate levels)
 - Predicted social media addiction score ~ 5.24 (moderate dependency)

Final Dataset Analysis: Categorical Profiling

- Demographic variables well balanced throughout (fair representation across student groups)
- AI tools widely adopted (various platforms and for various purposes)
- Psychological feature `predicted_burnout_risk` indicates that a substantial portion of students experience elevated stress levels
- Provide valuable contextual information for subsequent modelling and analysis

Final Dataset Overview

- **Study behavior** such as study hours and study-work hours appear **positively related** to final academic performance.
- **Social media usage** shows a **relationship with predicted addiction scores**, suggesting that heavy usage may correspond to stronger dependency indicators.
- **Wellbeing and burnout indicators** introduce an additional behavioral dimension. **Students with higher burnout risk may show greater variability in academic performance.**
- The integration of predicted psychological features enables the dataset to capture **mental health and behavioural factors** that were not directly measured in the original DS1 dataset.
- Motivate **subsequent stage of analysis** (predictive modelling of academic outcomes).